

Venkatesh Rapolu

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Summary

Entry-level **Java Backend Engineer** experienced in building scalable REST APIs and microservices using **Spring Boot, Spring Security, Redis, Docker, and Kubernetes**. Delivered production-grade systems handling **250+ requests/sec with 0% error rate** and **sub-30ms P90 latency**. Strong problem-solving foundation with **400+ DSA problems** solved on LeetCode and Codeforces.

Experience

Freelance Full-Stack Developer

Aug 2025 – Oct 2025

Acess Quiz SaaS Platform – Contract (2 Months)

Architected and delivered a **production-ready SaaS platform** independently, handling backend, frontend, payments, and deployment. Built secure **REST APIs** using **Spring Boot, Spring Security, JPA, Actuator, MongoDB, and Redis** for quiz creation, scheduling, evaluation, and leaderboards. Developed frontend using **React + Tailwind CSS**; integrated **Clerk authentication** with student ID verification and **Razorpay payment gateway** for subscription handling. **Dockerized** services and deployed on **VPS (Hostinger)**; completed full system in 1 month and provided maintenance, feature updates, and production support for 1 additional month.

Projects

Local Vendor Marketplace Platform (2026) [GitHub](#) — *Spring Boot, Spring Security, JWT, MySQL, Redis, Docker*

Built a **role-based backend system** with **USER/VENDOR/ADMIN RBAC** using **Spring Security & JWT**; implemented **Haversine geo-location vendor discovery** and **Redis caching** on hot endpoints; integrated **Cloudinary** for scalable image storage. Load tested with 20 concurrent users: achieved **250+ req/sec, 0% error rate**, and **sub-30ms P90 latency**. Distroless Docker build reduced image size from **600MB to 284MB – a 52% reduction**.

Voice Agent – AI Appointment Assistant (2025) [GitHub](#) [Live](#) — *Python, LangChain, Whisper, Firebase, React*

Built a **voice-driven AI assistant** for clinic patient intake using **Python, LangChain, and Whisper** for accurate speech-to-text processing. Designed **structured conversation flows** to capture symptoms and auto-generate appointment details; used **Firebase** for real-time storage, authentication, and cross-session sync. Reduced manual booking effort for clinic staff by approximately **30%**; deployed live on Vercel.

Uber-Like Ride Booking – Microservices (2025) [GitHub](#) — *Spring Boot, Spring Security, Microservices, REST APIs, Redis*

Architected a **cloud-native microservices ride platform** with 5 independent services: **AuthService, EntityService, ReviewService, LocationService**, and **ServerDiscovery**. Implemented **Redis in LocationService** for geospatial driver lookup and real-time proximity matching to reduce database load and improve response time. Validated full end-to-end ride booking lifecycle across services using REST-based inter-service communication.

Technical Skills

Languages: Java, C++, SQL, JavaScript

Backend: Spring Boot, Spring MVC, Spring Data JPA, Spring Security (JWT), REST APIs, gRPC

DevOps: Docker, Kubernetes, Jenkins CI/CD, GitHub Actions, GitLab CI, Linux (Bash)

Databases: MySQL, PostgreSQL, MongoDB, Redis (Caching, Geospatial)

Testing & Tools: JUnit, Mockito, Postman (Load Testing), Git, Kafka, Maven, Gradle

Achievements & Certifications

- **Winner – PU Code Hackathon 3.0:** Top 10 / 210 teams; built scalable backend system with real-time features [\[Certificate\]](#)
- **Finalist – Odoo Hackathon 2025:** Selected from 19,000+ participants; developed production-ready SaaS backend [\[Certificate\]](#)
- **Finalist – Vadodara Hackathon 5.0:** Top 45 / 400+ teams; built sign language translation application
- **MyOnsite Healthcare Hackathon:** Built AI-powered voice-based patient intake system [\[Certificate\]](#)
- **Postman API Fundamentals Student Expert:** Validated API design, testing, and request lifecycle expertise [\[Credential\]](#)
- **400+ DSA Problems:** Solved across LeetCode & Codeforces demonstrating strong algorithmic foundation

Education

Parul Institute of Technology

B.Tech in Computer Science & Engineering

2022 – 2026

CGPA: 8.39